

SCOPE OF WORK: DAIKIBO FACTORY TELEMETRY ANALYSIS

Data Analyst: Yana Dhanewar

Client/Sponsor: Daikibo Industrial

Stakeholders: Global Operations Manager, Technical Lead

Purpose:

The goal of this analysis is to evaluate one month (May 2021) of telemetry data from four global manufacturing hubs (Tokyo–Meiyo, Osaka–Seiko, Berlin, and Shenzhen). By identifying patterns in machine “unhealthy” signals, we can pinpoint specific locations and equipment types causing the most downtime to optimize maintenance schedules and reduce operational waste.

Scope / Major Project Activities:

Activity	Description
Data Ingestion	Importing and parsing the provided JSON telemetry dataset into Tableau.
Feature Engineering	Creating a calculated field [Unhealthy] using logic: IF [Status] = "unhealthy" THEN 10 ELSE 0 END to quantify downtime.
Exploratory Data Analysis (EDA)	Analyzing downtime distribution across 4 global factories and 9 machine types.
Interactive Visualization	Developing a dynamic dashboard with filter actions to allow drill-down from factory-level to machine-level diagnostics.

This project does not include:

- **Real-time Data Streaming:** The analysis is limited to the static JSON telemetry dataset provided for May 2021 and does not involve setting up live data connections to factory hardware.

- **Predictive Maintenance Modeling:** While the dashboard identifies current failure hotspots, this scope does not include building Machine Learning models to predict future equipment failures.
- **On-site Hardware Inspection:** This is a pure data diagnostic project; it does not involve physical testing or mechanical repair of the industrial machines (Laser Cutters, CNC, etc.) at the factory locations.

Deliverables:

Deliverable	Description/ Details
Cleaned Dataset Connection	A Tableau-ready connection to the unified JSON telemetry data.
Executive Dashboard	An interactive "Daikibo Operational Performance Overview" featuring downtime hotspots.
Root Cause Analysis	Identification of the specific factory and machine type driving failure rates.

Schedule Overview / Major Milestones:

Milestone	Expected Completion Date	Description/Details
<i>Data Readiness</i>	<i>Completed</i>	<i>Completion of JSON import and initial data schema validation.</i>
<i>Logic Implementation</i>	<i>Completed</i>	<i>Finalization of calculated fields and downtime weighting.</i>
<i>Dashboard Design</i>	<i>Completed</i>	<i>Integration of Sheets 1 & 2 into a synchronized interactive view.</i>
<i>Final Submission</i>	<i>TODAY</i>	<i>Documentation upload to GitHub and Portfolio.</i>

*Estimated date for completion:

Projected Final Submission: February 15th, 2026.